



ADAPTIVE MULTI-MODE ENVIRONMENTAL MONITORING & EMERGENCY ALERT NETWORK

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ABSTRACT

ARIA is an intelligent safety system that continuously monitors environmental conditions and sends periodic heartbeat signals to ensure active operation. It uses multi-sensor validation to classify threats, issuing warning alerts for gas, smoke, or high temperature without triggering panic. A confirmed emergency (flame detection or manual trigger) activates immediate response mechanisms and sends alerts via Wi-Fi, Telegram, web dashboard, and Li-Fi communication. Once the hazard is resolved, the system sends all-clear signals and returns to normal monitoring, ensuring reliable and adaptive safety management.

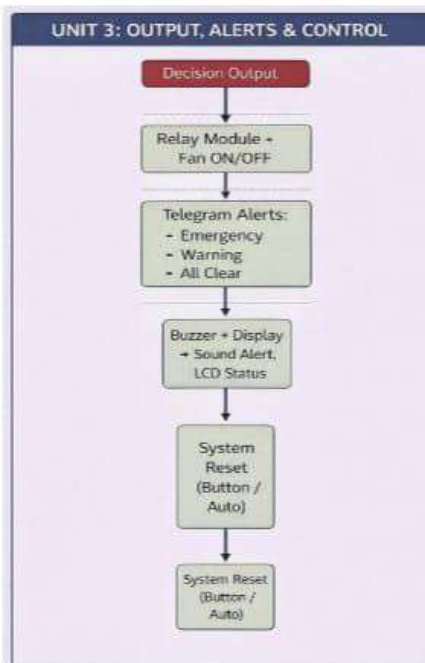
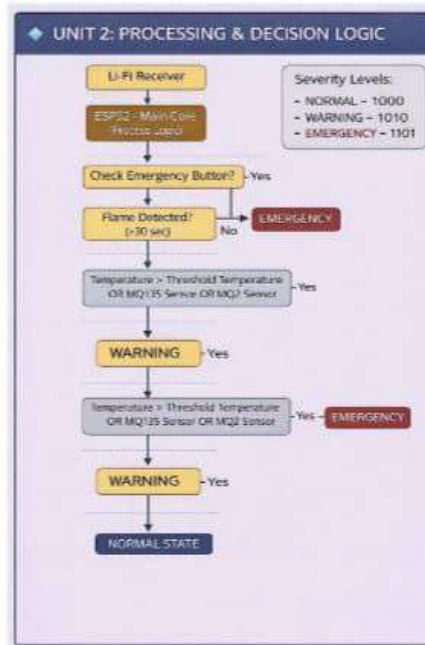
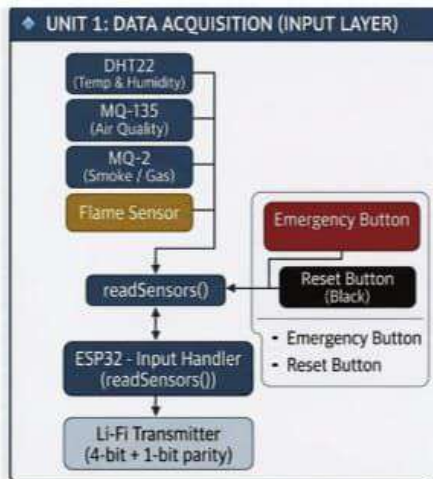
INTRODUCTION

Conventional safety systems rely on fixed thresholds and single-channel alerts, leading to false alarms and unreliable responses. ARIA addresses this with a structured multi-stage system, adaptive learning, and multi-sensor validation for accurate threat detection. Its reliability is enhanced through redundant communication via Wi-Fi, Telegram, web dashboards, and Li-Fi, making it ideal for sensitive environments.

OBJECTIVES

- Monitor temperature, humidity, air quality, smoke, and fire in real time
- Implement intelligent severity classification (Normal, Warning, Emergency)
- Enable multi-mode communication via Wi-Fi and Li-Fi for reliable alerts
- Provide real-time alert generation with automatic emergency response using relay systems

APPROACH



The system continuously monitors environmental parameters using sensors such as DHT22, MQ-135, MQ-2, and a flame sensor.

Data is processed by the ESP32 using an adaptive multi-sensor algorithm to classify conditions as Normal, Warning, or Emergency.

Early anomalies trigger Warning states, while Emergency is activated only upon flame detection or manual input, reducing false alarms.

Based on severity, the system activates buzzers and relay-controlled devices for response.

Alerts are transmitted via Wi-Fi to a web dashboard and sent through Telegram for real-time monitoring.

A Li-Fi module enables communication in RF-restricted environments by transmitting data through a laser diode.

An LDR at the receiver captures the signal, which is amplified and processed into visual or audible alerts.

RESULTS AND DISCUSSIONS

The proposed ESP32-based system successfully monitored environmental parameters such as temperature, humidity, air quality, smoke, and flame in real time.

The severity classification model efficiently categorized conditions into Normal, Warning, and Emergency states with fast response.

A hybrid communication system using Wi-Fi, Telegram, web dashboard, and Li-Fi ensured reliable and continuous alert transmission.

The relay module was accurately triggered during emergency conditions, demonstrating effective response capability.

Multi-sensor validation minimized false alarms, improving overall system reliability.

Additionally, its low power consumption and scalable design make it suitable for smart buildings, industries, and critical safety environments.

CONCLUSIONS AND FUTURE SCOPE

The system using the ESP32 enables real-time environmental monitoring and reliable multi-mode communication. It ensures fast detection and automatic emergency response using a relay module. The design improves safety, efficiency, and system reliability in critical environments. Future enhancements include IoT cloud integration and intelligent prediction using machine learning.

ACKNOWLEDGEMENTS

Our Sincere thanks to the faculty and head of the department, ECE, Principal and Management of Vasavi College of Engineering(A), Ibrahimbagh, Hyderabad for extending constant support and giving permission to access the hardware, software tools.

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